

Problem Description

You are given a weighted graph with N vertices and M edges. Find the total weight of its **maximum** spanning tree.

Constraints

- $1 \leq T \leq 20$
- $1 \leq N \leq 5000$
- $1 \leq M \leq 100\,000$
- $1 \leq c \leq 10\,000$

Input

The first line contains one integer T denoting the number of test cases. Each test case starts with a line containing **2** space-separated integers: N and M . Each of the following M lines contain description of one edge: three different space-separated integers: a , b and c . a and b are different and from 1 to N each and denote numbers of vertices that are connected by this edge. c denotes weight of this edge.

Output

For each test case output one integer - the total weight of its **maximum** spanning tree.

▼ Logical Test Cases

Test Case 1

INPUT (STDIN)

1
3 3
1 2 2
2 3 3
1 3 4

EXPECTED OUTPUT

7

Test Case 2

INPUT (STDIN)

1
3 3
1 2 3
2 3 4
1 3 5

EXPECTED OUTPUT

9

▼ Mandatory Test Cases

Test Case 1

KEYWORD

int printheap(int N)

▼ Complexity Test Cases

Test Case 1

CYCLOMATIC COMPLEXITY

2

Test Case 2

TOKEN COUNT

715

Test Case 3

NLOC

110